In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("C:\\Users\\Sanskruti\\Downloads\\zomato.csv\\zomato.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	url	address	name	online_order	book_table	rate	votes	phone	
0	https://www.zomato.com/bangalore/jalsa-banasha...	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n+91 9743772233	B
1	https://www.zomato.com/bangalore/spice-elephan...	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	B
2	https://www.zomato.com/SanchurroBangalore?cont...	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	B
3	https://www.zomato.com/bangalore/addhuri-udupi...	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	B
4	https://www.zomato.com/bangalore/grand-village...	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8/5	166	+91 8026612447\r\n+91 9901210005	Ba

In [4]:
```
df.shape
```

Out[4]:
```
(51717, 17)
```

In [5]:
```
df.columns
```

Out[5]:
```
Index(['url', 'address', 'name', 'online_order', 'book_table', 'rate', 'votes',
       'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
       'approx_cost(for two people)', 'reviews_list', 'menu_item',
       'listed_in(type)', 'listed_in(city)'],
      dtype='str')
```

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 51717 entries, 0 to 51716
Data columns (total 17 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51717 non-null  str
 1   address                      51717 non-null  str
 2   name                         51717 non-null  str
 3   online_order                 51717 non-null  str
 4   book_table                   51717 non-null  str
 5   rate                         43942 non-null  str
 6   votes                        51717 non-null  int64
 7   phone                        50509 non-null  str
 8   location                     51696 non-null  str
 9   rest_type                    51490 non-null  str
 10  dish_liked                   23639 non-null  str
 11  cuisines                     51672 non-null  str
 12  approx_cost(for two people)  51371 non-null  str
 13  reviews_list                 51717 non-null  str
 14  menu_item                    51717 non-null  str
 15  listed_in(type)              51717 non-null  str
 16  listed_in(city)              51717 non-null  str
dtypes: int64(1), str(16)
memory usage: 552.1 MB
```

In [7]:
```
df.describe()
```

Out[7]:

	votes
count	51717.000000
mean	283.697527
std	803.838853
min	0.000000
25%	7.000000
50%	41.000000
75%	198.000000
max	16832.000000

In [8]:
```
df.drop(['url','phone','dish_liked','reviews_list','menu_item'], axis=1, inplace=True)
```

In [9]:
```
df.rename(columns={'approx_cost(for two people)': 'cost_for_two'}, inplace=True)
```

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
address               0
name                  0
online_order          0
book_table            0
rate               7775
votes                 0
location             21
rest_type           227
cuisines             45
cost_for_two        346
listed_in(type)       0
listed_in(city)       0
dtype: int64
```

In [11]:
```
df.dropna(inplace=True)
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
address            0
name               0
online_order       0
book_table         0
rate               0
votes              0
location           0
rest_type          0
cuisines           0
cost_for_two       0
listed_in(type)    0
listed_in(city)    0
dtype: int64
```

In [13]:
```
df.drop_duplicates(inplace=True)
```

In [14]:
```
df = df[df['rate'].notna()]
df = df[~df['rate'].isin(['NEW','-'])]
df['rate'] = df['rate'].str[:3].astype(float)
```

In [15]:
```
df['cost_for_two'] = df['cost_for_two'].str.replace(',', '').astype(float)
```

In [16]:
```
df.head()
```

Out[16]:

	address	name	online_order	book_table	rate	votes	location	rest_type	cuisines	cost_for_two	listed_in(type)
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800.0	Buffet
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	Chinese, North Indian, Thai	800.0	Buffet
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	Cafe, Mexican, Italian	800.0	Buffet
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	South Indian, North Indian	300.0	Buffet
4	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	North Indian, Rajasthani	600.0	Buffet

In [17]:
```
df['cost_per_person'] = df['cost_for_two'] / 2
```

In [18]:
```
def category(x):
    if x >= 4:
        return 'Excellent'
    elif x >= 3:
        return 'Good'
    else:
        return 'Average'
df['rating_category'] = df['rate'].apply(category)
```

In [19]:
```
df[['rate', 'rating_category','cost_per_person']].head(10)
```

Out[19]:

	rate	rating_category	cost_per_person
0	4.1	Excellent	400.0
1	4.1	Excellent	400.0
2	3.8	Good	400.0
3	3.7	Good	150.0
4	3.8	Good	300.0
5	3.8	Good	300.0
6	3.6	Good	400.0
7	4.6	Excellent	300.0
8	4.0	Excellent	350.0
9	4.2	Excellent	275.0

In [20]:
```
df.iloc[0]
```

Out[20]:
```
address            942, 21st Main Road, 2nd Stage, Banashankari, ...
name                                                           Jalsa
online_order                                                     Yes
book_table                                                       Yes
rate                                                             4.1
votes                                                            775
location                                                Banashankari
rest_type                                             Casual Dining
cuisines                             North Indian, Mughlai, Chinese
cost_for_two                                                   800.0
listed_in(type)                                               Buffet
listed_in(city)                                         Banashankari
cost_per_person                                                400.0
rating_category                                            Excellent
Name: 0, dtype: object
```

In [21]:
```
df['location'].value_counts().head(10)
```

Out[21]:
```
location
BTM                      3873
Koramangala 5th Block    2295
HSR                      1992
Indiranagar              1795
JP Nagar                 1710
Jayanagar                1634
Whitefield               1568
Marathahalli             1407
Bannerghatta Road        1226
Koramangala 7th Block    1053
Name: count, dtype: int64
```

In [22]:
```
df['cuisines'].value_counts().head(10)
```

Out[22]:
```
cuisines
North Indian                            2107
North Indian, Chinese                   1949
South Indian                            1231
Cafe                                     620
Bakery, Desserts                         613
Biryani                                  600
South Indian, North Indian, Chinese      561
Desserts                                 545
Fast Food                                513
Chinese                                  409
Name: count, dtype: int64
```

In [23]:
```
df.groupby('location')['rate'].mean().sort_values(ascending=False).head(10)
```

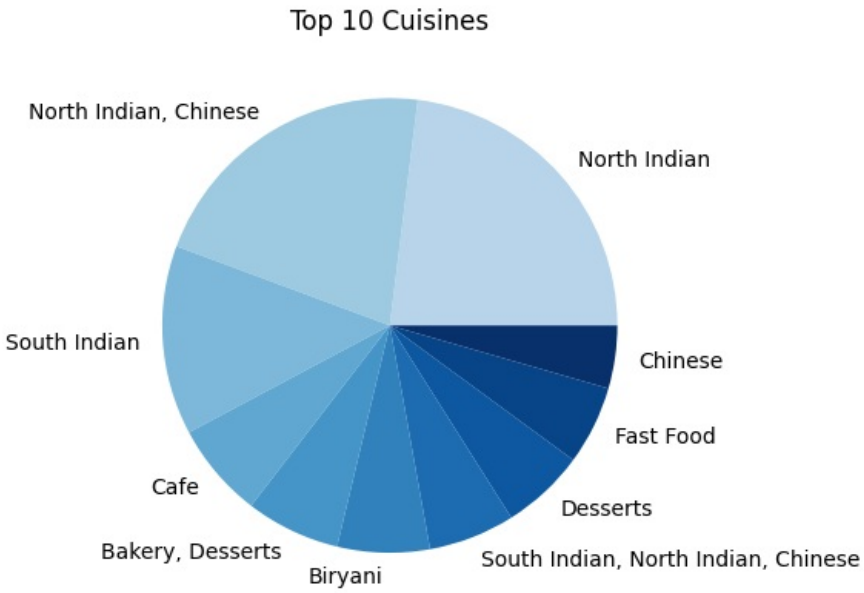
Out[23]:
```
location
Lavelle Road             4.140126
Koramangala 3rd Block    4.020526
St. Marks Road           4.017201
Koramangala 5th Block    4.006710
Church Street            3.990976
Sankey Road              3.965385
Koramangala 4th Block    3.918668
Cunningham Road          3.900633
Residency Road           3.862895
Koramangala 7th Block    3.851757
Name: rate, dtype: float64
```

In [25]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 41202 entries, 0 to 51716
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   address          41202 non-null  str
 1   name             41202 non-null  str
 2   online_order     41202 non-null  str
 3   book_table       41202 non-null  str
 4   rate             41202 non-null  float64
 5   votes            41202 non-null  int64
 6   location         41202 non-null  str
 7   rest_type        41202 non-null  str
 8   cuisines         41202 non-null  str
 9   cost_for_two     41202 non-null  float64
 10  listed_in(type)  41202 non-null  str
 11  listed_in(city)  41202 non-null  str
 12  cost_per_person  41202 non-null  float64
 13  rating_category  41202 non-null  str
dtypes: float64(3), int64(1), str(10)
memory usage: 11.2 MB
```

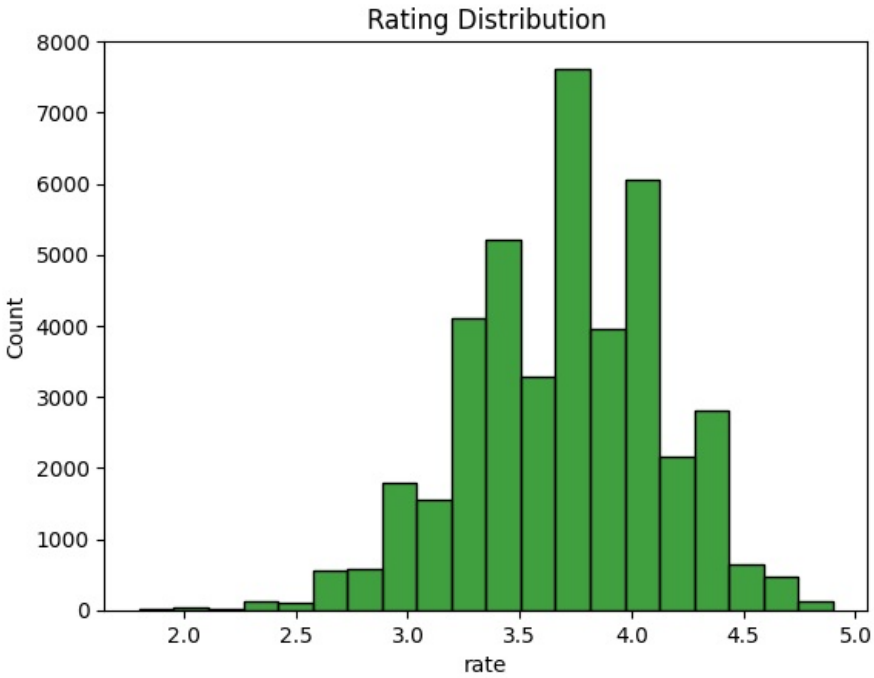
1. Top Cuisines

In [48]:
```
import matplotlib.pyplot as plt
import numpy as np
data = df['cuisines'].value_counts().head(10)
colors = plt.cm.Blues(np.linspace(0.3, 1, len(data)))
data.plot(
    kind='pie',
    colors=colors,)
plt.title("Top 10 Cuisines")
plt.ylabel("")
plt.show()
```



2. Rating Distribution

In [41]:
```
ax = sns.histplot(df['rate'], bins=20, color='green')
plt.title("Rating Distribution")
plt.show()
```

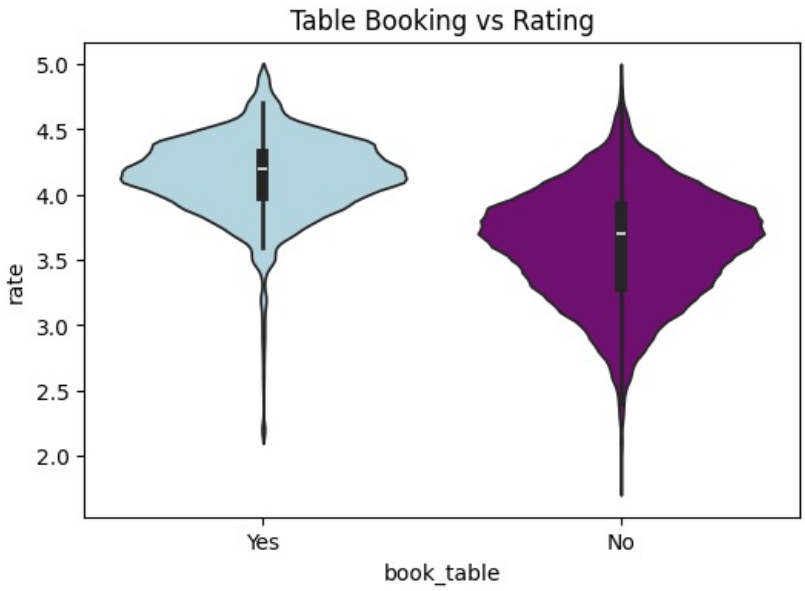


3. Online Order Vs Rating

In [55]:
```
ax = sns.boxplot(x='online_order', y='rate', data=df)
ax.set_facecolor('lightyellow')
plt.title("Online Order vs Rating")
plt.show()
```

4.Table Booking vs Rating

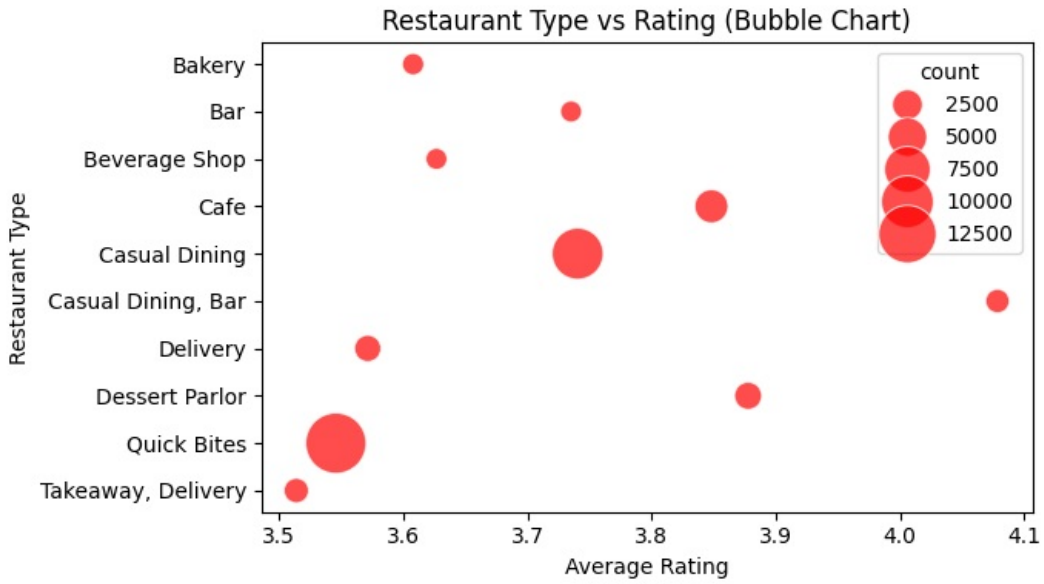
In [49]:
```
plt.figure(figsize=(6,4))
sns.violinplot(
    x='book_table',
    y='rate',
    hue='book_table',
    data=df,
    palette=['lightblue','purple'],
    legend=False)
plt.title("Table Booking vs Rating")
plt.show()
```



5.Restaurant Type vs Rating

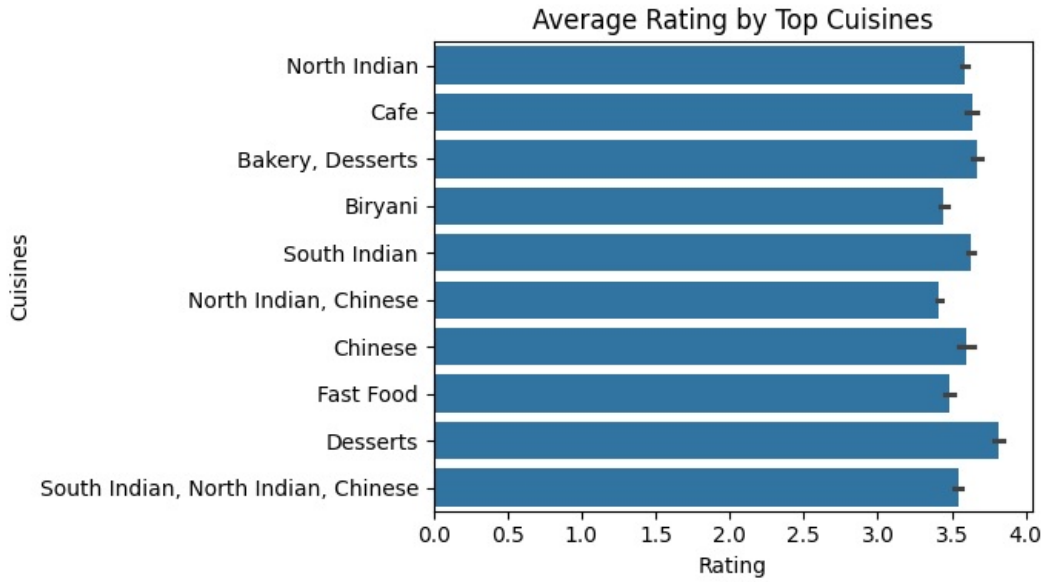
In [53]:
```
top_types = df['rest_type'].value_counts().head(10).index
avg_data = df[df['rest_type'].isin(top_types)].groupby('rest_type').agg({
    'rate':'mean',
    'rest_type':'count'
}).rename(columns={'rest_type':'count'}).reset_index()
plt.figure(figsize=(7,4))
sns.scatterplot(
    x='rate',
    y='rest_type',
    size='count',
    data=avg_data,
    sizes=(100, 800),
```

```
    alpha=0.7,
    color='red'
)
plt.title("Restaurant Type vs Rating (Bubble Chart)")
plt.xlabel("Average Rating")
plt.ylabel("Restaurant Type")
plt.tight_layout()
plt.show()
```



6.Rating by Top Cuisines

In [57]:
```
top_cuisines = df['cuisines'].value_counts().head(10).index
data = df[df['cuisines'].isin(top_cuisines)]
plt.figure(figsize=(7,4))
sns.barplot(x='rate', y='cuisines', data=data)
plt.title("Average Rating by Top Cuisines")
plt.xlabel("Rating")
plt.ylabel("Cuisines")
plt.tight_layout()
plt.show()
```



In [ ]: